A STEP-BY-STEP GUIDE

Excel Full Course Notes

From the interface to dashboards, one clear page at a time

1. What Are These Notes?

These notes walk through Excel from the interface to dashboards. This volume holds Modules 1 to 4 (pages 3 to 14). Every page follows the same pattern: what the topic is, the syntax or key shortcuts, worked examples, when to use it, important points, common mistakes and a Data Analyst tip.

2. Contents

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3. How to Read These Notes

Style	Meaning
Bold text	A key term to remember
`=SUM(A1:A5)`	A formula or key you can type
Important Things to Remember	The points most likely to matter in practice and in exams
Common Mistakes	Errors learners often make, with the fix
Data Analyst Tip	A habit that makes your analysis more reliable

Excel Basics

Module 1 | Know your way around before you calculate

1. What is Excel?

Excel is a **spreadsheet** program from Microsoft. It is used to store, calculate, analyse and visualise data. Everything lives in a grid made of **rows** and **columns**. Rows run across the sheet and are numbered (1, 2, 3...). Columns run down the sheet and are lettered (A, B, C...).

2. Key Terms

Term	Meaning	Example
Workbook	The whole Excel file	`Sales.xlsx`
Worksheet	One sheet (tab) inside a workbook	`Sheet1`
Cell	One box where a row and a column meet	`B3`
Range	A group of cells (the colon means "to")	`A1:C10`
Active cell	The cell with the thick green border, where you type	`Where you type`

3. Simple Example

	A	B
1	Item	Qty
2	Pen	10
3	Book	5
4	**Total**	**15**

Formula: `=SUM(B2:B3)`
B2 is a **cell**. B2:B3 is a **range** (B2 "to" B3). Excel adds 10 + 5.
Result: 15

4. Size Limits (Good to Know)

Item	Limit
Rows per sheet	1,048,576
Columns per sheet	16,384 (A to XFD)
Characters in one cell	32,767
Sheets in a workbook	Limited only by memory

5. When to Use It

- Store lists such as customers, orders or marks in rows and columns
- Calculate totals, averages and percentages automatically
- Analyse data with sorting, filtering, PivotTables and charts
- Share results as tables, charts or dashboards

6. Important Things to Remember

- A cell address is the column letter first, then the row number: B3, not 3B.
- A range uses a colon: `A1:C10` means every cell from A1 to C10.
- The active cell is where anything you type will go.
- A worksheet is one tab. A workbook is the whole file with all its tabs.

7. Common Mistakes

Mistake: Writing the row number before the column letter, such as 3B. **Fix:** Always write the letter first: B3.

Mistake: Typing into the wrong cell because you did not notice which cell was active. **Fix:** Look for the thick green border, or check the Name Box before typing.

8. Data Analyst Tip

Know the limits before you import data. A sheet holds 1,048,576 rows, and a larger CSV file will be cut off when opened in Excel. For bigger data, load it with Power Query (Module 14).

Interface Tour & File Types

Module 1 | Know your way around before you calculate

1. What is the Interface?

The interface is everything you see around the grid: the toolbars, the Ribbon, the bars that show cell contents and the tabs at the bottom. Knowing where each part is saves a lot of time.

2. Parts of the Screen

Part	What it does
Quick Access Toolbar	Save, Undo and Redo at the very top
Ribbon	Tabs that hold all commands: Home, Insert, Page Layout, Formulas, Data, Review, View
Name Box	Shows the active cell address. Type an address here to jump there
Formula Bar	Shows what is really inside the active cell
Sheet tabs	Bottom left. Click + to add a new sheet
Status Bar	Bottom. Shows Sum, Average and Count of the selected cells

3. What Each Ribbon Tab is For

Tab	Use it for
Home	Fonts, alignment, number format, styles, sort and filter
Insert	Tables, PivotTables, charts, sparklines, shapes
Page Layout	Margins, orientation, print area, themes
Formulas	Function library, Name Manager, auditing tools
Data	Import data, sort, filter, validation, what-if
Review	Spell check, notes, protect sheet
View	Freeze panes, zoom, gridlines, page break preview

4. File Types

Type	What it is
`.xlsx`	Normal workbook
`.xlsm`	Workbook with macros
`.csv`	Plain text, values separated by commas, no formatting
`.xls`	Old Excel 97-2003 format
`.pdf`	Export to share a read-only copy

5. Simple Example: Status Bar

	A	B
1	Student	Marks
2	Rahim	78
3	Karim	85
4	Hasan	92
5	Nabil	66
6	Sara	79

Select B2:B6 and look at the Status Bar. It shows **Average: 80, Count: 5, Sum: 400**. No formula is needed.

6. When to Use It

- Check a total or average quickly without typing a formula
- Jump to a far-away cell using the Name Box
- Find the right command by knowing its Ribbon tab
- Choose the correct file type before saving or sharing

7. Important Things to Remember

- The cell shows a result, but the Formula Bar shows what is really stored.
- Save as .xlsm only when the workbook contains macros. A normal .xlsx cannot hold them.
- A .csv file keeps values only. Formulas, colours and extra sheets are lost.

8. Common Mistakes

Mistake: Saving a workbook with formatting and several sheets as .csv. **Fix:** Save as .xlsx and export a .csv copy only when another program needs it.

9. Data Analyst Tip

Select a column of numbers and check the Status Bar. If Count is lower than the number of rows, some cells are blank or stored as text, so investigate before you analyse.

Navigation Shortcuts

Module 1 | Know your way around before you calculate

1. What are Navigation Shortcuts?

Navigation shortcuts move or select around a sheet using only the keyboard. In a sheet with thousands of rows they are far faster than scrolling with the mouse.

2. Shortcuts

Shortcut	What it does
`Ctrl + Arrow key`	Jump to the edge of the data block
`Ctrl + Home`	Go to cell A1
`Ctrl + End`	Go to the last used cell
`Ctrl + Space`	Select the whole column
`Shift + Space`	Select the whole row
`Ctrl + A`	Select all (press twice inside a table)
`Ctrl + PgDn / PgUp`	Next / previous sheet
`F5 or Ctrl + G`	Go To dialog

3. Simple Example

	A	B
1	Name	Marks
2	Rahim	70
3	Karim	80
4	Hasan	60
5		
6		
7	Sara	90

Select B2 and press `Ctrl + Down Arrow`. The cursor jumps to B4, the end of the block. Press it again and it jumps to B7, the next filled cell.

4. Another Example

Press `F5` (or `Ctrl + G`), type `D500` and press Enter. Excel jumps straight to cell D500.

Press `Ctrl + Home` to return to A1, and `Ctrl + End` to see the last cell Excel considers used.

5. When to Use It

- Reach the bottom of a long list in one keystroke
- Select an entire column or row before formatting it
- Move between sheets without the mouse
- Check whether stray cells exist far from your data

6. Important Things to Remember

- `Ctrl + Arrow` stops at blank cells inside the data. Press it again to jump to the next filled cell.
- `Ctrl + End` goes to the last **used** cell. This can be further than your visible data if old formatting is left behind.
- Inside a table, `Ctrl + A` first selects the table data. Press it a second time to select the whole sheet.
- Add `Shift` to any `Ctrl + Arrow` move to select cells along the way.

7. Common Mistakes

Mistake: Pressing Ctrl + Down Arrow and assuming the end of the data was reached, when a blank cell stopped the jump. **Fix:** Check for blank rows, or press Ctrl + End to confirm the real last cell.

Mistake: Scrolling with the mouse through thousands of rows. **Fix:** Use Ctrl + Arrow, Ctrl + Home and Ctrl + End.

8. Data Analyst Tip

Select a whole data column quickly with `Ctrl + Shift + Down Arrow` from its first cell. You then see the exact range to use in a formula, and any blank cell that breaks the selection stands out immediately.

Entering Data & AutoFill

Module 2 | Type fast, format clean

1. What is Data Entry?

Data entry is how values get into cells. Excel decides how to treat what you type, so it helps to know the types of data and the keys that make typing faster.

2. Types of Data

Type	Examples	Notes
Text	Names, codes	Left aligned by default
Numbers	1250, 3.5	Right aligned by default
Dates and times	01-01-2026	Stored as numbers, shown as dates
Logical	TRUE, FALSE	Results of comparisons
Errors	#N/A, #DIV/0!	Shown when a formula cannot work

3. Entering and Editing

Key	What it does
`Enter / Tab / Esc`	Enter moves down, Tab moves right, Esc cancels typing
`F2`	Edits the cell without double clicking
`Alt + Enter`	Starts a new line inside the same cell
`Ctrl + ;`	Enters today's date
`Ctrl + Shift + ;`	Enters the current time
`'007`	Starts with an apostrophe to keep leading zeros as text
`Ctrl + D / Ctrl + R`	Fill the cell above downwards / fill from the left

4. Simple Example: AutoFill

	A
1	Value
2	10
3	20
4	30
5	40

Type 10 and 20 first so Excel learns the pattern. Select both and drag the **fill handle** (the small square at the corner of the active cell). Excel continues 30, 40 and so on. It also continues 1, 2, 3 or Jan, Feb, Mar or Mon, Tue, Wed.

5. Another Example: Flash Fill

	A	B
1	Full Name	First Name
2	Rahul Sharma	Rahul
3	Priya Verma	Priya
4	Amit Das	Amit

Type `Rahul` next to `Rahul Sharma`. In the next row press `Ctrl + E`. **Flash Fill** copies the pattern down the column.

6. When to Use It

- Number rows, dates or months without typing each one
- Split or join text such as first and last names with Flash Fill
- Keep codes like 007 exactly as typed
- Stamp the current date or time into a sheet

7. Important Things to Remember

- Dates and times are numbers underneath. Only the format makes them look like dates.
- Starting with an apostrophe makes the entry text, so leading zeros are kept.
- Flash Fill works only when Excel can see a clear pattern in your first example.

8. Common Mistakes

Mistake: Typing 007 and finding Excel shows 7. **Fix:** Type '007 (apostrophe first) or format the cell as Text before typing.

Mistake: Dragging the fill handle on a single number and getting copies instead of a series. **Fix:** Type two values first, select both, then drag.

9. Data Analyst Tip

Use `Ctrl + ;` for a date that stays fixed. The `TODAY()` function changes every day, which can silently change old reports.

Number Formats & Layout

Module 2 | Type fast, format clean

1. What is Number Formatting?

Number formatting changes how a value **looks** without changing the value itself. Press `Ctrl + 1` to open Format Cells. The real value stays the same inside the cell, so always check the Formula Bar.

2. Number Formats

Format	Typed 1234.5 shows as
General	1234.5
Number (2 decimals)	1234.50
Currency	$1,234.50
Percentage (typed 0.25)	25%
Date (typed 46023)	01-01-2026
Custom `#,##0`	1,235
Custom `0.00 "kg"`	1234.50 kg

3. Simple Example

	A	B
1	Typed	Shows as
2	0.25	25%
3	1234.5	1234.50 kg
4	**=A3*2**	**2469**

Formula: `=A3*2`
Cell A3 shows `1234.50 kg` but still holds 1234.5. Excel uses the real value. **Result: 2469**

4. Alignment and Layout

Feature	What it does
Wrap Text	Keeps long text inside the cell by growing the row height
Center Across Selection	Centres text over several cells without merging
AutoFit	Double click a column border to fit the width

5. Insert, Delete and Hide Shortcuts

Shortcut	What it does
`Ctrl + Shift + +`	Inserts cells
`Ctrl + -`	Deletes cells
`Ctrl + 9`	Hides a row
`Ctrl + 0`	Hides a column

6. When to Use It

- Show currency, percentages or units without changing the numbers
- Make long headings readable with Wrap Text
- Tidy a sheet by hiding helper rows and columns
- Control how many decimals appear in reports

7. Important Things to Remember

- Formatting only changes how a value looks. The real value is shown in the Formula Bar.
- **Merge & Center** looks neat but breaks sorting and filtering. Prefer **Center Across Selection** (Format Cells > Alignment).
- Typing 0.25 in a Percentage cell shows 25%.

8. Common Mistakes

Mistake: Assuming a number is rounded because it shows 2 decimals. **Fix:** Use `=ROUND(A1,2)` when the stored value must change.

Mistake: Merging header cells and then being unable to sort the table. **Fix:** Use Center Across Selection instead.

9. Data Analyst Tip

If a total looks off by a small amount, click the cells and compare what the Formula Bar holds with what the cell shows. Hidden decimals are a common cause.

Paste Special & Find and Replace

Module 2 | Type fast, format clean

1. What are They?

Paste Special (`Ctrl + Alt + V`) lets you paste only part of what you copied, such as values or formats. **Find and Replace** (`Ctrl + F` and `Ctrl + H`) searches your data and changes it in bulk.

2. Paste Special Options

Option	Why use it
Values	Keep the result, drop the formula
Formulas	Copy formulas without formatting
Formats	Copy only the look
Column widths	Make widths match
Transpose	Swap rows and columns
Multiply / Add	Change a whole range by a number in one go

3. Simple Example: Multiply

	A	B
1	Price	After x 1.18
2	100	118
3	200	236
4	300	354

Copy a cell holding `1.18`. Select the prices, choose Paste Special > **Multiply**. Each price is multiplied by 1.18 (18% tax) in one step.

4. Wildcards in Find

Wildcard	Meaning	Finds
*	Any characters	`Sh*` finds Sharma, Shah
?	One character	`R?hul` finds Rahul
~*	A real star	`~*` finds the symbol *

5. Find and Replace Basics

- `Ctrl + F` finds and `Ctrl + H` replaces.
- Click **Options** to search the whole workbook or inside formulas.
- To find the wildcard characters themselves, put a tilde before them: `~*` finds a real star.

6. When to Use It

- Convert formulas to fixed numbers with Paste Special > Values
- Flip a table from rows to columns with Transpose
- Apply a price rise or tax to a whole range at once
- Clean data by replacing a wrong word or symbol everywhere

7. Important Things to Remember

- Values removes the formula, so the number will no longer update.
- Wildcards: * means any characters and ? means one character.
- Find and Replace searches only the current sheet unless you change Within to Workbook.

8. Common Mistakes

Mistake: Clicking Replace All without checking what will change. **Fix:** Use Find All first to review the matches, or replace one at a time.

Mistake: Copying formulas to another file and getting broken references. **Fix:** Paste as Values when the numbers only need to be kept.

9. Data Analyst Tip

Keep a backup copy of the sheet before a large Replace All or Multiply paste. Both are easy to run and hard to undo after you save and close.

Formula Basics & Operators

Module 3 | Operators, references and errors

1. What is a Formula?

A formula is an instruction that tells Excel to calculate something. **Every formula starts with an equals sign.** Without it, Excel treats your entry as plain text.

2. Syntax

=A2+B2

This adds the values in A2 and B2. If A2 changes, the answer updates automatically.

Operator	Meaning	Example
+ -	Add, subtract	`=10-4 gives 6`
* /	Multiply, divide	`=6*3 gives 18`
^	Power	`=2^3 gives 8`
&	Join text	`="Hi "&A1`
%	Percent	`=50%*200 gives 100`
= > < >= <= <>	Comparisons	`=A1>=40 gives TRUE`

3. Simple Example

	A	B	C
1	Price	Tax	Total
2	**100**	**18**	**118**

Formula in C2: `=A2+B2`
Excel adds 100 + 18. **Result: 118**. Change A2 to 200 and C2 updates to 218.

4. Order of Operations

Excel follows **BODMAS**: Brackets, then Powers, then Multiply / Divide (left to right), then Add / Subtract.

Formula	Result	Why
`=2+3*4`	14	3*4 happens first
`=(2+3)*4`	20	Brackets go first

5. When to Use It

- Add, subtract, multiply or divide values that may change later
- Join text, such as a greeting and a name
- Compare values to get TRUE or FALSE
- Build totals, taxes and discounts that update automatically

6. Important Things to Remember

- Forgetting the equals sign makes Excel show your formula as text.
- Use brackets to control the order. `=(2+3)*4` gives 20, but `=2+3*4` gives 14.
- Text inside a formula needs double quotes, such as `="Hi "&A1`.
- A comparison such as `=A1>=40` returns TRUE or FALSE, not a number.

7. Common Mistakes

Mistake: Typing 2+3*4 without the equals sign and seeing the text instead of 14. **Fix:** Start every formula with =.

Mistake: Expecting (2+3)*4 when typing =2+3*4. **Fix:** Add brackets around the part that must be calculated first.

8. Data Analyst Tip

Never type numbers that already sit in cells into a formula. Point to the cell, as in `=A2+B2`, so the result updates when the data changes.

Cell References & Named Ranges

Module 3 | Operators, references and errors

1. What are Cell References?

A cell reference tells a formula where to look. What happens to a reference when you copy the formula depends on its type.

Type	Looks like	When you copy it
Relative	`A1`	Changes with position
Absolute	`$A$1`	Never changes
Mixed (column fixed)	`$A1`	Row changes only
Mixed (row fixed)	`A$1`	Column changes only

2. Syntax

Press `F4` while editing a reference to cycle: `A1` > `$A$1` > `A$1` > `$A1`.

3. Simple Example: Absolute

	A	B	C
1	Item	Price	Tax
2	Pen	100	18
3	Book	200	36
4	**Bag**	**300**	**54**

Tax rate 18% is in E1. Formula in C2: `=B2*$E$1`
The $ locks E1, so copying down still points to the same rate.

4. Another Example: Relative

Copy `=A2*B2` from C2 to C3 and Excel writes `=A3*B3` for you. That is a **relative reference** at work.

5. Other Sheets and Files

Where	Example
Another sheet	`=Sheet2!A1`
Name with spaces	`='Jan Sales'!A1`
Another workbook	`='[Budget.xlsx]Sheet1'!A1`

6. Named Ranges

- Select the cells, type a name in the Name Box and press Enter.
- Then use it: `=SUM(Sales)` instead of `=SUM(C2:C100)`.
- `Ctrl + F3` opens Name Manager to edit or delete names.
- Names cannot have spaces and cannot look like a cell address.

7. When to Use It

- Apply one tax rate, discount or target to a whole column
- Pull numbers from other sheets or workbooks
- Make long formulas readable with named ranges

8. Important Things to Remember

- Relative (A1) changes when copied, absolute ($A$1) never changes.
- Mixed references lock only the column ($A1) or only the row (A$1).
- Copying a formula only adjusts relative parts of its references.

9. Common Mistakes

Mistake: Copying `=B2*E1` down so the rate moves to E2, E3 and returns 0 or wrong values. **Fix:** Lock the rate with `=B2*$E$1` (press F4).

Mistake: Naming a range with a space or a name like AB12. **Fix:** Use a name such as `Sales_Q1`.

10. Data Analyst Tip

Put assumptions such as tax rate or target in their own labelled cells and point to them with absolute references. You can then change one cell and update the whole model.

Common Errors & Auditing Tools

Module 3 | Operators, references and errors

1. What are Formula Errors?

When Excel cannot complete a calculation it shows an error code instead of a result. Each code points to a different cause, so reading it tells you where to look.

2. Common Errors and Fixes

Error	Meaning	Usual fix
`#####`	Column too narrow	Widen the column
`#DIV/0!`	Divided by zero or blank	Check divisor, or use IFERROR
`#N/A`	Value not found	Check lookup value and spelling
`#NAME?`	Unknown function or name	Fix the typo
`#REF!`	Reference was deleted	Rebuild the reference
`#VALUE!`	Wrong data type	Numbers stored as text?
`#NUM!`	Invalid number	Check the arguments
`#SPILL!`	Spill range is blocked	Clear the cells in the way

3. Simple Example: #DIV/0!

	A	B	C
1	Sales	Orders	Average
2	**500**	**0**	**#DIV/0!**

Formula in C2: `=A2/B2`
Dividing by 0 gives an error. Fix: `=IFERROR(A2/B2,0)` returns **0**.

4. Another Example: #VALUE!

	A	B	C
1	Qty	Note	Total
2	**10**	**ten**	**#VALUE!**

Formula in C2: `=A2+B2`
B2 holds text, so Excel cannot add it. Replace the text with a number.

5. Auditing Tools (Formulas Tab)

Tool	What it does
Trace Precedents / Dependents	Draw arrows to related cells
Evaluate Formula	Walks through a formula step by step
Ctrl + ` (key under Esc)	Shows all formulas on the sheet
F9 on part of a formula	Shows that part's value. Press Esc afterwards, do not press Enter
Calculation Options	If numbers do not update, it may be set to Manual. Press F9 to recalculate

6. When to Use It

- Find out why a formula returns an error
- Check which cells feed a result and which cells depend on it
- Review a whole sheet of formulas at once

7. Important Things to Remember

- `#####` is not a real error. The column is simply too narrow to show the number.
- Pressing Enter after F9 replaces that part of the formula with its value. Press Esc instead.
- If numbers do not update, check Formulas > Calculation Options.

8. Common Mistakes

Mistake: Wrapping every formula in IFERROR so errors disappear. **Fix:** Find the cause first. Use IFERROR only where an error is expected.

Mistake: Pressing Enter after F9 and overwriting part of the formula. **Fix:** Press Esc to cancel.

9. Data Analyst Tip

Read the error code before you change anything. The code tells you the kind of problem, and Evaluate Formula then shows the exact step where it goes wrong.

Functions: Anatomy & Statistics

Module 4 | The everyday toolkit

1. What is a Function?

A function is a ready-made formula with a name. You give it values to work on, and it returns a result. It saves you from writing long calculations by hand.

2. Syntax

```
=FUNCTION(argument1, argument2, ...)
```

Name first, then brackets. Arguments are separated by commas. Type `=` and the first letters, then press `Tab` to accept the suggestion. `Shift + F3` opens the Insert Function box. A colon means a range (`A1:A5`). A comma means separate cells (`A1,A5`).

3. Simple Example

	A	B
1	Student	Marks
2	Rahim	78
3	Karim	85
4	Hasan	92
5	Nabil	66
6	Sara	79

Marks are in B2:B6: 78, 85, 92, 66, 79. Every formula below uses this range.

4. What Each Formula Returns

Formula	Result	What it does
`=SUM(B2:B6)`	400	Adds all numbers
`=AVERAGE(B2:B6)`	80	Mean value
`=MAX(B2:B6)`	92	Largest
`=MIN(B2:B6)`	66	Smallest
`=MEDIAN(B2:B6)`	79	Middle value
`=LARGE(B2:B6,2)`	85	2nd largest
`=SMALL(B2:B6,2)`	78	2nd smallest
`=RANK.EQ(85,B2:B6)`	2	Rank, largest = 1

5. When to Use It

- Total sales, marks or expenses with SUM
- Find the average, highest and lowest values in a list
- Find the top or bottom 2nd, 3rd... values with LARGE and SMALL
- Rank students, products or salespeople

6. Important Things to Remember

- AVERAGE is the SUM divided by how many numbers there are: 400 ÷ 5 = 80.
- MEDIAN is the middle value once the numbers are in order: 66, 78, 79, 85, 92.
- RANK.EQ gives rank 1 to the largest value by default.
- Every opening bracket needs a closing bracket.

7. Common Mistakes

Mistake: Using a comma when you meant a range, such as `=SUM(B2,B6)` instead of `=SUM(B2:B6)`. **Fix:** Use a colon for a continuous range and commas for separate cells.

Mistake: Including the header row or a Total row in the range. **Fix:** Select only the data rows.

8. Data Analyst Tip

Compare the AVERAGE with the MEDIAN. Here they are close (80 and 79), so the data is fairly even. A big gap means a few extreme values are pulling the average.

Counting, Rounding & SUBTOTAL

Module 4 | The everyday toolkit

1. What are These Functions?

These helpers count cells, tidy numbers and total only the rows you can see. They are used in almost every report.

2. Counting Functions

Function	Counts
`COUNT`	Cells that contain numbers
`COUNTA`	Cells that are not empty
`COUNTBLANK`	Empty cells
`ROWS / COLUMNS`	Number of rows / columns in a range

3. Simple Example: Counting

	A	B
1	Name	Marks
2	Rahim	70
3	Karim	80
4	Hasan	
5	Nabil	Absent
6	Sara	90

Results for B2:B6

Formula	Result
`=COUNT(B2:B6)`	3
`=COUNTA(B2:B6)`	4
`=COUNTBLANK(B2:B6)`	1

COUNT sees only the numbers 70, 80 and 90. COUNTA also counts the text Absent. COUNTBLANK counts the empty B4.

4. Rounding and Number Helpers

Formula	Result	Note
`=ROUND(12.456,2)`	12.46	Normal rounding
`=ROUNDUP(12.451,2)`	12.46	Always away from zero
`=ROUNDDOWN(12.459,2)`	12.45	Always toward zero
`=ROUND(1250,-2)`	1300	Negative = left of the point
`=INT(-7.2)`	-8	Rounds down to an integer
`=MOD(10,3)`	1	Remainder after dividing
`=ABS(-5)`	5	Removes the minus sign

5. SUBTOTAL: Totals That Respect Filters

`=SUBTOTAL(9, B2:B100)`

Code 9 = SUM, 1 = AVERAGE, 2 = COUNT, 3 = COUNTA, 4 = MAX, 5 = MIN. It only counts rows left visible after a filter. Use codes 101-111 to also ignore rows you hid by hand. For example, 109 is SUM that skips hidden rows.

6. When to Use It

- Count filled, empty or numeric cells in a column
- Round prices or percentages for reports
- Total only the filtered rows
- Test for even or odd numbers: `=MOD(A1,2)=0` is TRUE for even numbers

7. Important Things to Remember

- COUNT counts numbers only, COUNTA counts anything that is not empty.
- A negative digit in ROUND works left of the decimal point: `=ROUND(1250,-2)` gives 1300.
- SUM adds hidden and filtered rows too. SUBTOTAL skips filtered rows.

8. Common Mistakes

Mistake: Using SUM on a filtered list and getting the total of all rows. **Fix:** Use `=SUBTOTAL(9, range)`.

Mistake: Formatting to 2 decimals and expecting the stored number to be rounded. **Fix:** Use ROUND to change the real value.

9. Data Analyst Tip

Place `=COUNTA(B2:B6)` next to `=COUNT(B2:B6)`. If COUNT is smaller than COUNTA, some numbers are stored as text.

Percentages, PMT & Random Numbers

Module 4 | The everyday toolkit

1. What are They?

Percentage formulas answer share, growth, tax and discount questions. **PMT** calculates a regular loan payment. **RAND** and **RANDBETWEEN** create random numbers for tests and samples.

2. Syntax

```
=PMT(rate/12, years*12, -loan)
```

This gives the monthly payment. Keep the units matching: if payments are monthly, divide the yearly rate by 12 and multiply the years by 12.

Task	Formula	Example
Share of total	`=B2/SUM($B$2:$B$6)`	Format as %. 78 of 400 = 19.5%
Growth	`=New/Old-1`	From 120 to 150 gives 25%
Add 18% tax	`=B2*1.18`	100 becomes 118
Discount 10%	`=B2*(1-10%)`	100 becomes 90

3. Simple Example: Growth

	A	B	C
1	Old	New	Growth
2	**120**	**150**	**25%**

Formula in C2: `=B2/A2-1`
150 ÷ 120 = 1.25, minus 1 = 0.25. **Result: 25%**

4. Another Example: PMT

	A	B	C	D
1	Loan	Rate	Years	Payment
2	**500000**	**10%**	**5**	**10,623.52**

Formula in D2: `=PMT(B2/12,C2*12,-A2)`
A 5 year loan of 500,000 at 10% a year gives about **10,623.52** a month.

5. Random Numbers and Quick Tools

- `=RAND()` gives a decimal between 0 and 1.
- `=RANDBETWEEN(1,100)` gives a whole number between 1 and 100.
- Both change every time the sheet recalculates. Paste as values to freeze them.

6. When to Use It

- Work out each product's share of total sales
- Measure growth between two months or years
- Add tax or apply a discount to a price list
- Compare monthly loan payments before borrowing

7. Important Things to Remember

- Growth is New ÷ Old - 1, then format the cell as a percentage.
- Keep PMT units matching: monthly payments need a monthly rate and months of term.
- RAND and RANDBETWEEN recalculate whenever the sheet does.

8. Common Mistakes

Mistake: Using the yearly rate in PMT for monthly payments. **Fix:** Divide the rate by 12 and multiply the years by 12.

Mistake: Getting a negative payment from PMT. **Fix:** Put a minus sign before the loan amount: `-A2`.

Mistake: Forgetting to lock the total in a share formula and getting wrong results when copying down. **Fix:** Use `$B$2:$B$6` so the range does not move.

9. Data Analyst Tip

Check the total cost of a loan: 10,623.52 × 60 months = 637,411.20. That is 137,411.20 more than the 500,000 borrowed, which is the interest you pay.